B. Tech.

ELECTRICAL ENGINEERING

Syllabus of Minor Specialization

AY: 2021-22

Authored By: Charusat

SYLLABUS

for

MINOR SPECIALIZATION

2021-22

(Choice Based Credit System)

Faculty of Technology & Engineering

M & V Patel Department of Electrical Engineering

Bachelor of Technology Programme

(Electrical Engineering)

Chandubhai S Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of technology

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives

M & V Patel Department of Electrical Engineering

Vision

To become an internationally recognized, research intensive education entity, serving as a workforce developer for the nation by providing the best education in close collaboration with society.

Mission

To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.

THE PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1	**Preparation:** To prepare the students with sound academically background, intensive development of knowledge and skills, and with an uncompromising attitude to achieve excellence in the electrical engineering.
PEO 2	**Core Competence:** To build a broad engineering core competence and to enable the students to solve the problems systematically by providing them exposure to different dimensions of specializations.
PEO 3	**Learning Environment:** To provide conducive learning environment instilling sound foundation in mathematics, computing techniques, science and humanities along with professional courses for futuristic thinking in an increasingly technology dependent society.
PEO 4	**Professionalism:** To provide a platform to the students to plan and shape their career in the chosen field and to imbibe ethical values with thorough professionalism and encouragement towards entrepreneurial thinking and traits.

PROGRAM ARTICULATION MATRIX

(-Not Matching, 1-Low, 2-Medium, 3-High)

	PEO 1	PEO 2	PEO 3	PEO 4
To create world class technocrats with strong theoretical foundations and practical engineering skills who can contribute through the technological applications and research for the betterment of lives.	3	3	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO 1	**Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO 2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO 4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	**The engineer and society:** Apply reasoning obtained by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	**Individual and teamwork:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO 1	Apply logical, analytical and technical skills to different areas like electrical and electronic circuits, power electronics, power systems, electrical machines, PLC, renewable energy integration and its conservation.
PSO 2	Apply recent advances like Artificial intelligence, machine learning and optimization for design, simulation and analysis of electrical systems to provide solutions to real world problem.

Minor Specialization in **"Artificial Intelligence and Machine Learning"**

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
4	CE291	Python Programming Skills	2	4	6	4	30	70	50	50	**200**
5	CE391	Python For Data Analytics	3	2	5	4	30	70	25	25	**150**
6	CE392	Machine Learning Fundamentals	4	2	6	5	30	70	25	25	**150**
7	CE491	Deep Learning applications and AI	4	2	6	5	30	70	25	25	**150**
			13	**10**	**23**	**18**	**120**	**280**	**125**	**125**	**650**

Minor Specialization in **"Electrical Vehicle Systems"**

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
4	ME291	Basics of Electrical Vehicles	3	0	3	3	30	70	-	-	**100**
5	EE391	Control of Electric Motors For Vehicular Applications	4	2	6	5	30	70	25	25	**150**
6	EE392	EV Batteries and Charging System	4	2	6	5	30	70	25	25	**150**
7	EE491	Electric Vehicles in Smart Grid	4	2	6	5	30	70	25	25	**150**
			15	**6**	**21**	**18**	**120**	**280**	**75**	**75**	**550**

B. Tech. (Electrical Engineering) Programme

SYLLABI

Minor Specialization In Artificial Intelligence and Machine Learning

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE291: PYTHON PROGRAMMING SKILLS

B.TECH 4TH SEMESTER (Electrical)

INSTITUTE ELECTIVE - I

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	2	4	6	4
Marks	100	100	200	

A. **Objective of the Course:**

The objectives of the course are:

- To describe the core syntax and semantics of Python programming language.
- To illustrate the process of structuring the data using lists, dictionaries, tuples and sets.
- To introduce function - focussed programming paradigm through python.
- To infer the Object-oriented Programming concepts in Python.
- To indicate the use of regular expressions and built-in functions to navigate the file system.
- To train in the development of solutions using modular concepts.

B. **Outline of the course:**

Sr. No.	Title of the unit	Minimum number of hours
1.	Basics of Python	01
2.	Variables and Data Types	04
3.	Loops and Conditional Control	06
4.	Advanced Data Structures	04
5.	Functions	02
6.	Object Oriented Programming in python	08
7.	Reading and Writing Files	03
8.	Packages and Modular Programming	02

Total Hours (Theory): 30

Total Hours (Lab): 60

Total Hours: 90

C. Detailed Syllabus:

1.	**Basics of Python**	**01 Hour**	**03%**
1.1	Introduction to Python		
1.2	Installation and Environment Setup		
1.3	Simple Python Program Practice		
2.	**Variables and Data Types**	**04 Hours**	**13%**
2.1	Variables, Operators, Precedence and Associativity		
2.2	Data Types , Indentation, Comments, Reading Input, Print Output		
2.3	Memory Optimization		
2.4	Type Conversion		
3.	**Loops and Conditional Control**	**06 Hours**	**20%**
3.1	The if… Decision Control Flow Statement, The if…else Decision Control Flow Statement, The if…elif…else Decision Control Statement		
3.2	Nested if Statement, The while Loop, The for Loop,		
4.	**Advanced Data Structures**	**04 Hours**	**13%**
4.1	Lists- Basic List Operators, Replacing, Inserting, Removing an Element, Searching and Sorting Lists, Tuples.		
4.2	Dictionaries- Dictionary Literals, Adding and Removing Keys, Accessing and Replacing Values; Traversing Dictionaries.		
4.3	Sets, Strings, Files and their Libraries		
5.	**Functions**	**02 Hour**	**07%**
5.1	Defining a Function, Calling a Function		
5.2	Pass by Reference vs Value, Function Arguments, Required and Default Arguments, Keyword Arguments and Variable-Length Arguments		
5.3	The Anonymous Functions, The Return Statement		
5.4	Scope of Variables		
6.	**Object Oriented Programming in python**	**08 Hours**	**27%**
6.1	Object-Orientation Introduction, Class Definition, Class Objects,		

	Instance Objects, Method Objects		
6.2	Class and Instance Variables, Inheritance, Private Variables, The Polymorphism		
7.	**Reading and Writing Files**	**03 Hour**	**10%**
7.1	Reading Files, Parsing and Processing Files		
7.2	File storage, File operations (delete, copy)		
8.	**Packages and Modular Programming**	**02 Hour**	**07%**
8.1	Executing modules as Scripts, Import and Use of Built-in Modules, Accessing Third Party Packages and Modules		
8.2	Creating Custom Modules and Packages, Importing and Using Custom Modules		

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Practicals/Tutorials related to course content will be carried out in the laboratory.

E. Course Outcomes (COs):

At the end of the course the students will be able to:

CO1	Interpret the fundamental python syntax, semantics and fluent in the use of python control flow statements. (Understand)
CO2	Determine the methods to create and manipulate python programs by utilizing the data structures like lists, dictionaries, tuples and sets. (Apply)
CO3	Express proficiency in the handling of strings and functions. (Apply)
CO4	Articulate the Object-Oriented Programming concepts such as inheritance and

	polymorphism as used in Python. (Create)
CO5	Identify and use the commonly used operations involving file systems, and regular expressions (Create)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	-	-	-	1	-	1	2	-	-	2	-	-
CO2	-	1	-	2	2	3	2	-	-	-	2	-	2	-
CO3	1	-	2	1	3	2	3	2	2	2	-	2	-	3
CO4	-	1	2	2	3	3	2	3	-	3	3	-	-	-
CO5	1	2	-	2	3	3	2	2	2	-	3	3	3	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

F. Recommended Study Material:

❖ Text Books:

1. Nagar, Sandeep, "Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation", Apress, 2018.
2. Kenneth Lambert, "Fundamentals of Python: First Programs", ISBN-13: 978-1337560092, Cengage learning publishers, first edition, 2012.
3. Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
4. Reema Thareja, "Python Programming using Problem Solving Approach", ISBN-13:978-0-19- 948017-3, Oxford University Press, 2017.

❖ Reference Books:

1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, OREILLY, 2016
2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
3. Allen Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press, 2015

4. Mark Lutz "Learning Python", 4th Edition, O‟‟REILLY, 2016
5. Vamsi kurama, “Python programming: A modern approach”, ISBN-978-93-325-8752- 6, Pearson, 2018.
6. W. Chun, “Core python programming”, ISBN-13: 978-0132269933, Pearson, 2nd edition, 2016.

❖ **Web Materials:**

1. https://www.python.org/
2. http://www.diveintopython3.net/
3. https://developer.mozilla.org/en-US/docs/Learn/Server-side/Django
4. https://www.fullstackpython.com/django.html
5. https://codelabs.developers.google.com/
6. https://www.tutorialspoint.com/python/index.htm

❖ **Software:**

1. Python IDLE
2. Anaconda Python
3. PyCharm

LIST OF PRACTICALS

Practical No.	Name of Practical
1	Determination of transformation ratio, nominal ratio, phase angle error and ratio error for Potential Transformer using python programming.
2	Determination of transformation ratio, nominal ratio, phase angle error and ratio error for Current Transformer using python programming.
3	To perform the operations of various logic gates using python programming.
4	To find out the transfer function of the system with Block Diagram Reduction Method using python programming.
5	Perform the Nodal analysis on a given DC Resistive Circuit using python programming.
6	Perform the Mesh analysis on a given DC Resistive Circuit using python programming
7	Perform the Fourier Transform using python programming.
8	Perform the calculation of different types of Tariffs using python programming.
9	Perform the depreciation cost determining methods using python programming.
10	Calculate the cost of electrical energy using python programming

CE391: PYTHON FOR DATA ANALYTICS

B.TECH. 5^{TH} SEMESTER (Electrical)

INSTITUTE ELECTIVE - II

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Objectives of the Course:

The main objectives of the course are:

- To provide comprehensive knowledge of python programming paradigms required for Data Science
- To get familiar with the main elements of Python programming
- To understand the basic concepts of Python
- To become Proficient in using the Most Common Python Data Science Packages like Numpy, Pandas & Matplotlib
- To understanding how to manipulate and analyze uncurated datasets
- To execute the basic statistical analysis methods

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	**NumPy**	10
2	**Pandas**	12
3	**Matplotlib**	10
4	**Statistics for data analytics**	13

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

C. **Detailed Syllabus:**

1	**NumPy**	**10 Hours**	**22%**
1.1	NumPy Introduction		
1.2	Create NumPy Arrays		
1.3	Data Type Object, type		
1.4	Numerical Operations on NumPy Arrays		
1.5	Changing the Dimensions of Arrays		
1.6	Python, NumPy and Probability		
1.7	Creation of Synthetical Test Data		
1.8	Boolean Masking of Arrays		
1.9	Matrix Arithmetic		
1.10	Reading and Writing and arrays		
2	**Pandas**	**12 Hours**	**27%**
2.1	Pandas Data Frames		
2.2	Replacing Values in Data Frames and Series		
2.3	Pandas Data Files		
2.4	Dealing with NaN		
2.5	Binning Data		
2.6	Pandas Tutorial Continuation: multi-level indexing		
2.7	Data Visualization with Pandas and Python		
2.8	Python, Pandas and Timeseries		
2.9	Estimating the number of Corona Cases with Python and Pandas		
3	**Matplotlib**	**10Hours**	**22%**
3.1	Pyplot API		
3.2	Simple Plot		
3.3	Figure Class		
3.4	Axes Class		
3.5	Multiplot		
3.6	Subplots () Function		
3.7	Formatting Axes		
3.8	Setting Ticks and Tick Labels		

3.9	Bar Plot		
3.10	Histogram		
3.11	Pie Chart		
3.12	Scatter Plot		
3.13	Three-dimensional Plotting		
4	**Statistics for data analytics**	**13Hours**	**29%**
4.1	Mean, Median, Mode		
4.2	Standard Deviation, Variance		
4.3	Variable Distribution		
4.4	Normalization		
4.5	Statistical Modeling using Auto Regression and Auto		
4.6	Correlation		

D. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted as per pedagogy as a part of internal theory evaluation.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.
- Experiments/Tutorials related to course content will be carried out in the laboratory.

E. Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand mechanisms for missing data and analytic implications (Understand)
CO2	Build and code a Graphical User Interface (GUI) to run a program
CO3	Understand and use python data science libraries as a tool for data analytics (Apply)
CO4	Implement numerical programming, data handling and visualization through NumPy, Pandas and Matplot-lib modules. (Create)
CO5	Analyze the significance of python program development environment by working on real world examples (Create)

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	1	-	-	-	1	-	-	2	-	-	-	-	-
CO2	-	1	-	2	2	3	2	-	-	-	-	-	-	-
CO3	1	-	2	-	2	2	3	2	2	2	2	2	-	-
CO4	-	1	2	2	2	3	3	2	-	3	3	3	2	-
CO5	1	2	-	2	2	3	3	2	2	-	3	3	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. **Recommended Study Material:**

Text Books:

1) William McKinney, “ Python for Data Analysis: Data Wrangling with Pandas, NumPy, and Ipython” O’Reilly Media, Inc.,
2) Nagar, Sandeep, “Introduction to Python for Engineers and Scientists - Open Source Solutions for Numerical Computation”, Apress, 2018.
3) Jesus Rogel-Salazar, “Data Science and Analytics with Python”, CRC Press, Taylor & Francis Group, LLC

Reference Books:

1) Magnus Lie Hetland, "Beginning Python from Novice to Professional", Third Edition, Apress, 2017
2) Mark Lutz "Learning Python", 4th Edition, O‟REILLY, 2016

Web Materials:

1) https://nptel.ac.in/courses/106/106/106106212/
2) https://matplotlib.org/gallery/index.html
3) https://realpython.com/python-matplotlib-guide/
4) https://www.tutorialspoint.com/python/index.htm

LIST OF PRACTICALS

Practical No.	Name of Practical
1	Write Python programs to understand control structures.
2	Write Python programs to understand list and tuples.
3	Use conditional statements and loops in Python programs.
4	Write python programs to create functions and use functions in the program.
5	Import module and use it in Python programs.
6	Write python program to plot data using PyPlot.

CE392: MACHINE LEARNING FUNDAMENTALS
B.TECH 6TH SEMESTER (Electrical)
INSTUTUTE ELECTIVE - III

Credits and Hours:

Teaching Scheme	Theory	Practical/Tutorial	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

G. Objective of the course:

The main objectives for offering the course Artificial Intelligence are:

- To learn about the most effective machine learning techniques, and gain practice implementing them.
- To able to effectively use the common neural network "tricks", including initialization, dropout regularization, Batch normalization, gradient checking,
- To understand industry best-practices for building deep learning applications.
- To learn how to quickly and powerfully apply these techniques to new problems.

H. Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to AI and Machine Learning	12
2.	Supervised Learning	16
3.	Evaluation Metrics	06
4.	Unsupervised Learning	10
5.	Neural Network	16

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

I. Detailed Syllabus:

1.	**Introduction to AI and Machine Learning**	**12 Hours**	**20%**
1.1	Environment Setup. Introduction to Machine Learning Pipeline		
2.	**Supervised Learning**	**16 Hours**	**27%**
2.1	Regression, Simple Linear regression Least Square method, Simple Linear regression Using gradient Descent, Multiple Linear regression: Application,		
2.2	Classification, Logistic Regression, Log Loss. Sigmoid, K-Nearest Neighbour		
3.	**Evaluation Metrics**	**06 Hours**	**10%**
3.1	Regression (SSE, MSE, RMSE)		
3.2	Classification (Accuracy, precision, Recall, F1 Score, Confusion Matrix)		
4.	**Unsupervised Learning**	**10 Hours**	**16%**
4.1	Anomaly Detection		
5.	**Neural Network**	**16 Hours**	**27%**
5.1	Perceptron, Perceptron Algorithm, Single Layer NN, Back Propagation, Tuning, Application (Classification, Regression)		

J. Instructional Method and Pedagogy:

At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.

- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams will be conducted as per pedagogy as a part of internal theory evaluation.
- Assignments based on course content will be given to the students at the end of each unit/topic and will be evaluated at regular interval.
- Surprise tests/Quizzes/Seminar will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

- Experiments/Tutorials related to course content will be carried out in the laboratory.

K. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To solve difficult and complex problem of computer science using AI techniques (Apply).
CO2	To select any R&D field related to application of AI (Understand, Apply).
CO3	To understand soft computing and machine learning courses (Understand).
CO4	To develop software solution as per need of today's IT edge which requires high automation and less human intervention (Understand, Create, Apply)
CO5	To demonstrate working knowledge in Python in order to write and explore more sophisticated Python programs (Apply).
CO6	To apply knowledge representation, reasoning, and machine learning techniques to real-world problems (Apply).

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	2	-	-	2	-	-	1	2	2
CO2	3	3	2	2	2	-	-	-	-	-	-	-	-	1
CO3	2	-	-		-	-	-	-	-	-	-	-	1	1
CO4	3	3	3	2	1	1	-	-	-	-	-	-	1	1
CO5	2	2	2	2	2	1	-	-	-	-	-	2	-	1
CO6	3	3	3	3	2	2	-	-	-	-	-	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

L. Recommended Study Material:

❖ Text Books:

1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077

2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 2004

❖ Reference Books:

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification, Second Edition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of statistical learning", Springer, 2001.
4. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: An introduction", MIT Press, 1998.

❖ Web Materials:

1. https://www.youtube.com/watch?v=foHSmB48rY&list=PLKvX2d3IUq586Ic9gIhZj6ubpWV-OJfl4
2. https://www.youtube.com/watch?v=CS4cs9xVecg&list=PLkDaE6sCZn6EcXTbcX1uRg2_u4xOEky0
3. https://www.youtube.com/watch?v=UzxYlbK2c7E
4. https://www.youtube.com/watch?v=fgtUFzxNztA
5. http://www-formal.stanford.edu/jmc/whatisai/whatisai.html
6. https://www.webopedia.com/TERM/A/artificial_intelligence.html
7. https://en.wikipedia.org/wiki/Artificial_intelligence

❖ Software

1. Scikit Learn
2. PyTorch
3. TensorFlow
4. Colab
5. Keras

LIST OF PRACTICALS

Practical No.	Name of Practical
1	Application in Structural Engineering – Damage Detection, Damage

	Localizations, Damage Quantification, real time Structural Health Monitoring.
2	Application in Geotechnical Engineering – Soil Moisture Estimation, Soil Classification using Machine Learning
3	Application in Construction Management – Site Layout using Machine Learning, Machine Learning for preliminary estimation, Predicting change of rate of material, risk prediction and management systems, construction activity monitoring system, Site safety monitoring by Machine Learning, Project schedule optimization
4	Concrete mix design using Machine Learning
5	Material Recognition and design

CE491: DEEP LEARNING APPLICATIONS AND AI

B.TECH 7TH SEMESTER (Electrical)

INSTITUTE ELECTIVE - IV

Credits and Hours:

Teaching Scheme	Theory	Tutorial/Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Objective of the Course:

The main objectives of the course are:

- To make students aware about Deep Learning and Artificial Intelligence applications
- To learn about the most effective AI techniques, and gain practice implementing them
- To learn how to quickly and powerfully apply these techniques to new problems
- To make students aware about field where AI can be applied in civil engineering problems

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	**Introduction**	10
2	**Deep neural network**	12
3	**Improving deep neural network**	10
4	**Convolution Neural Network**	10
5	**Sequence models**	10
6	**Structuring Deep learning project**	08

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 90

C. Detailed Syllabus:

1	**Introduction**	**10 Hours**	16%
1.1	Introduction: deep learning		
1.2	Tools for deep learning		
1.3	Shallow and deep neural network		
1.4	Tensor flow		
2	**Deep Neural Network**	**12 Hours**	20%
2.1	Forward propagation		
2.2	Backpropagation		
2.3	Application: Load Forecasting		
3	**Improving Deep Neural Network**	**10 Hours**	17%
3.1	Learning rate, Activation Function		
3.2	Number of Layers and Neurons		
3.3	Overfitting and Underfitting Test		
4	**Convolution Neural Network**	**10 Hours**	17%
4.1	Convolution operation		
4.2	Padding and strides		
4.3	Activation functions		
4.4	Pooling and dropout layers		
5	**Sequence models**	**10 Hours**	17%
5.1	Time series and sequence data		
5.2	Modelling data for sequence learning		
5.3	Recurrent neural network		
5.4	Long short-term memory network		
6	**Structuring Deep learning project**	**08 Hours**	13%

D. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and practical session which carries 10 Marks and 5 marks weightage respectively.

- Two internal exams will be conducted and average of the same will be converted to equivalent of 15 Marks as a part of internal theory evaluation.
- Assignment/Surprise tests/Quizzes/Seminar will be conducted which carries 5 Marks as a part of internal theory evaluation.
- The course includes a practical session where students have an opportunity to build an appreciation for the concepts being taught in lectures.

E. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To understand fundamental of Deep Learning and Artificial Intelligence (Understand).
CO2	To solve difficult and complex problem of computer science using AI techniques (Understand, Apply).
CO3	To identify fields where AI can be applied (Understand, Apply).

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	1	1	1	2	-	-	-	-	-	-	1	1	1
CO2	2	2	2	2	1	1	-	-	-	-	-	1	1	1
CO3	2	1	1	2	1	1	-	-	-	-	-	1	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

F. Recommended Study Material:

Reference Books:

1. Stuart Russell, Artificial Intelligence – A Modern Approach (3 Edition), Pearson, U.K., 2016.
2. Tariq Rashid, Make your own neural network, CreateSpace Independent Publishing Platform, U.S., 2016.

3. Ian Goodfellow, Yoshua Bengio and Aaron Courville, Deep Learning, The MIT Press, Cambridge, England, 2017.
4. Seth Weidman, Deep Learning from scratch, O'Reilly media, U.S.A., 2019.

Web Materials:

1. https://www.udemy.com/course/artificial-intelligence-for-civil-engineers-part-1/
2. https://www.udemy.com/course/artificial-intelligence-for-civil-engineers-part-2/

LIST OF PRACTICALS

Practical No.	Name of Practical
1	Application of hybrid artificial intelligence approaches for hydraulic problems
2	Using artificial neural networks for estimation of the energy consumption obtained by dams and hydropower plants
3	Swarm intelligence for optimum weight of the structures
4	Metaheuristic optimization algorithm in design optimization of 2D and 3D steel or concrete structure
5	Application of fuzzy logic in civil engineering problems
6	Optimization model for single- and multi-objective solutions for construction management

B. Tech. (Electrical Engineering) Programme

SYLLABI

Minor Specialization In Electrical Vehicle Systems

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME291: BASICS OF ELECTRICAL VEHICLES

4th Semester and 2nd Year (Level II)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-1
Institute Elective

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	-	3	3
Marks	100	00	100	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	-	-	100

C. Course Objectives:

1. To make students understand the basic concepts, requirements and working of various components of automobile.
2. To enable students with working of different automobile systems and subsystems.
3. To aware students about recent technologies in automobile engineering and its working.
4. To introduce with basic idea about how automobile systems are developed.

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to Electric Vehicles	14
2	Automotive Design	14
3	Design of Electric and Hybrid Electric Vehicles	15
4	Regulation and Standardization of Vehicles	02

Total hours (Theory)	:	45 Hrs
Total hours (Lab)	:	00 Hrs
Total hours	:	45 Hrs

E. Detailed Syllabus:

1	**Introduction to Electric Vehicles**	**14Hours**	**31.11%**
1.1	I.C Engine powered Vehicles Vs Electrical Vehicles (EVs), How EV works, Types of EVs, Hybrid Electric Drive-train		
1.2	Tractive effort in normal driving, Energy consumption Concept of Hybrid Electric Drive Trains, Architecture of Hybrid Electric Drive Trains, Series Hybrid Electric Drive Trains, Parallel hybrid electric drive trains, Electric Propulsion unit.		
2	**Automotive Design**	**14Hours**	**31.11%**
2.1	Major parts of automobiles, Chassis, Frame, Body of automobile, Transmission system, Steering and Breaking systems, Suspension, Accessories.		
3	**Design of Electric and Hybrid Electric Vehicles**	**15Hours**	**33.34%**
3.1	Series Hybrid Electric Drive Train Design: Operating patterns, control strategies, Sizing of major components, power rating of traction motor, power rating of engine/generator, design of PPS		
3.2	Parallel Hybrid Electric Drive Train Design: Control strategies of parallel hybrid drive train, design of engine power capacity, design of electric motor drive capacity, transmission design, energy storage design		
4	**Regulation and Standardization of Vehicles**	**02Hours**	**4.44%**
4.1	Motor vehicle act, registration of motor vehicles, driving license, control of traffic, insurance against third party, claims for compensation, traffic signs, central motor vehicle		

rules, vehicle safety standards and regulations, classification and definition of vehicles, enforcement of emission norms, duties of surveyor.

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand needs and working of various systems in automobiles.
CO2:	Practically identify different automotive systems and subsystems.
CO3:	Practically identify different automotive components.
CO4:	Understand market and businesses of automobile industry.
CO5:	This course will give the students some awareness to recent trends and research areas in Automobiles.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	1	1	-	-	-	-	-	1	-	-	1	-	-
CO3:	3	1	1	-	-	-	-	-	1	-	-	2	-	-
CO4:	-	-	-	-	-	-	-	-	-	-	2	1	-	-

CO5:	1	1	1	1	-	-	-	-	1	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realiase the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

I. Recommended Study Material:

Text Books:

1. Singh Kirpal, "Automobile Engineering" Volume I & II, Standard Pub.& Dist.

2. Gupta K. M. “Automobile Engineering” Volume I & II, Umesh Pub.

3. Gupta R. B. “Automobile Engineering”, Satya Prakashan.

4. Giri N. K., “Automobile Technology”, Khanna Pub.

Reference Books:

1. Crouse W., “Automotive Mechanics”, Tata Mc Graw Hill.

2. Narang G. B. S. “Automobile Engineering”, Khanna Pub.

3. John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley

Web Material:

1. http://nptel.iitm.ac.in

2. http://www.sae.org/pubs/automotive/

3. https://www.springer.com/journal/12239/

Other Material:

1. International Journal of Automotive Technology (www.ijat.net/)
2. International Journal of Automotive Technology

EE391: CONTROL OF ELECTRIC MOTORS FOR VEHICULAR APPLICATIONS

5^{th} Semester and 3^{rd} Year (Level III)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-2
Institute Elective

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

- To understand the different electrical machines used for vehicle application.
- To design the DC motor control with sensor as well as without sensor.
- To understand the control of induction motor with respect to electric vehicle .
- To understand the control strategies for PM motors.
- To understand the brushless motor drive and magnet less motor drive

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Electrical Machines for Vehicular Applications	15
2	DC Motor Dynamics & Control	10

3	Induction Motor Dynamics & Control	11
4	Special Electric Motor Dynamics & Control	24

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

E. **Detailed Syllabus:**

1 Electric Machines for Vehicular Applications 15 Hours 25 %

Overview of conventional ac and dc motors, special electric machines such as permanent magnet machines, switched reluctance machines, linear motors, machine and engine rating, requirements, torque – speed characteristics and speed control, cooling of electric machines, testing of machines and testing standards

2 DC Motor Dynamics & Control 10 Hours 16.66 %

Various schemes of closed loop control using sensors, Sensor less operation of drive, DC motor drive using PWM rectifiers.

3 Induction Motor Dynamics & Control 11 Hours 18.34%

Principles of soft starting, Loss minimization, Various schemes of closed loop control, Direct and indirect field oriented control of induction motor, Direct torque control.

4 Special Electrical Motors Dynamics & Control 24 Hours 40 %

- Switched Reluctance Motor Drive: Structure of SR Machines, Principle of SR Machines, Converter configurations, Modified converter topology, Control strategies, Sensor and sensor less control of SRM, Reduction of torque pulsation.
- Stator-PM Motor Drives : Doubly-Salient PM Motor Drives, Flux-Reversal PM Motor Drives, Flux-Switching PM Motor Drives, Hybrid-Excited PM Motor Drives Flux-Mnemonic PM Motor Drives, Design Criteria of Stator-PM Motor Drives for EVs, Application, Advantages and Limitations for EVs.
- Brushless Motor Drive: Fundamentals of permanent magnet motors, Control strategies, Vector control of the Permanent Magnet Motor drive, Direct torque

control of Permanent Magnet Motor drive, Sensor less control, Reduction of torque pulsations, Parameter sensitivity of drive.

- Advanced Magnet less Motor Drives: Introduction of Advanced Magnet less technology, Synchronous Reluctance Motor Drives, Doubly-Salient DC Motor Drives, Flux-Switching DC Motor Drives

F. **Revised Bloom's Taxonomy**

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. **Course Outcomes (COs):**

Upon successful completion of this course, a student would be able to

CO1:	Understand the different electrical machines used for vehicle application.
CO2:	Design the DC motor control with sensor as well as without sensor.
CO3:	Understand the control of induction motor with respect to electric vehicle
CO4:	Understand the control strategies for PM motors
CO5:	Understand the brushless motor drive and magnet less motor drive

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	1	-	-	-	-	-	-	-	-	1	-	-
CO2:	3	2	3	1	2	2	-	-	1	-	-	2	1	-

CO3:	3	2	1	1	1	-	-	-	-	-	-	1	1	-
CO4:	3	2	1	1	1	-	-	-	-	-	-	1	1	-
CO5:	3	1	-	-	1	-	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- In lab session, the student will be going through the various motors drive control strategies and simulations which will help to make some demo drives of the various motors used for electrical vehicles.

I. Recommended Study Material:

Text Books:	
1.	Mehrdad Ehsani, Yimin Gao and Ali Emadi, "Modern Electric, Hybrid Electricand Fuel cell vehicles: Fundamentals, Theory and Design", CRC Press, 2004.
2.	Electric Vehicle Machines and Drives Design, Analysis and Application by K. T. Chau, Wiley-IEEE Press; 1st edition

3.	Advanced Electric Drive Vehicles (Energy, Power Electronics, and Machines) by Ali Emadi , CRC Press; 1st edition (9 December 2014)
Reference Books:	
1.	Electric Vehicle Technology Explained by James Larminie & John Lowry, Wiley
Web Material:	
1.	https://nptel.ac.in/courses/108/106/108106170/
2.	https://onlinecourses.nptel.ac.in/noc21_ee112/
3.	https://nptel.ac.in/courses/108/102/108102121/

EE392: EV BATTERIES AND CHARGING SYSTEM
6th Semester and 3rd Year (Level III)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-3
Institute Elective

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

1. To understand the basics of battery and physics of battery.
2. To understand and select the proper battery for electric vehicle.
3. To analyze the different charging strategies for EVs.
4. To analyze and design the converters for battery charging.
5. To understand about the fuel cell and design battery management system

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Basics of Batteries	06
2	EV Batteries	08

3	EV Charging	10
4	Power Electronic Converter for Battery Charging	15
5	Battery Management System	15
6	Fuel cells	06

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

E. Detailed Syllabus:

1 Basics of Batteries 06 Hours 10%

Basics – Types, Parameters – Capacity, Discharge rate, State of charge, state of Discharge, Depth of Discharge, Technical characteristics, Battery pack Design, Properties of Batteries, Cell and battery voltages, Charge (or Amphour) capacity, Energy stored, Energy density, Specific power, Amphour (or charge) efficiency, Energy efficiency, Self-discharge rates, Battery geometry, Battery temperature, heating and cooling needs, Battery life and number of deep cycles

2 EV Batteries 08 Hours 13.33%

Lead Acid Batteries : Lead acid battery basics, Special characteristics of lead acid batteries, Battery life and maintenance, Battery charging, **Nickel-based Batteries :** Introduction, Nickel cadmium, Nickel metal hydride batteries, **Lithium Batteries :**Introduction, The lithium polymer battery, The lithium ion battery ,Recent development in EV batteries, Modeling of Batteries

3 EV Charging 13 Hours 21.67%

Charging Infrastructure: Domestic Charging Infrastructure, Public Charging Infrastructure, Normal Charging Station, Occasional Charging Station, Fast Charging Station, Battery Swapping Station, Move-and-charge zone. Battery Chargers: Charge equalisation, Conductive (Basic charger circuits, Microprocessor based charger circuit. Arrangement of an off-board conductive charger, Standard power levels of conductive chargers, Inductive

(Principle of inductive charging, Soft-switching power converter for inductive charging), Battery indication methods

4 **Power Electronic Converter for Battery Charging** 12 Hours 20 %

Charging methods for battery, Termination methods, charging from grid, The Z-converter, Isolated bidirectional DC-DC converter, Design of Z-converter for battery charging, High-frequency transformer based isolated charger topology, Transformer less topology

5 **Battery Management System** 15 Hours 25%

Introduction and BMS functionality, Battery pack topology, BMS Functionality, Voltage Sensing, Temperature Sensing, Current Sensing, BMS Functionality, High-voltage contactor control, Isolation sensing, Thermal control, Protection, Communication Interface, Range estimation, State-of charge estimation, Cell total energy and cell total power, Battery state of charge estimation (SOC), voltage-based methods to estimate SOC, Model-based state estimation, Battery Health Estimation, Lithium-ion aging: Negative electrode, Lithium ion aging: Positive electrode, Cell Balancing, Causes of imbalance, Circuits for balancing

6 **Fuel cells** 06 Hours 10%

Thermodynamics of fuel cells; free energy change and cell potentials; effects of temperature and pressure on cell potential; energy conversion efficiency; factors affecting conversion efficiency; polarization losses; important types of fuel cells (hydrogen-oxygen, organic compounds-oxygen, carbon or carbon monoxide-air, nitrogen compounds-air); Equivalent circuits of fuel cells.

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the basics of battery and physics of battery.
CO2:	Understand and practically select the proper battery for electric vehicle.
CO3:	Analyze the different charging strategies for EVs.
CO4:	Design the converters for battery charging.
CO5:	Understand about the fuel cell and design battery management system

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	-	-	-	-	-	-	-	-	-1	-	-	-	-
CO2:	3	1	1	-	1	-	-	-	-	-	-	1	-	-
CO3:	3	1	1	1	2	-	-	-	1	-	-	1	1	-
CO4:	3	3	3	1	2	-	-	-	1	-	1	1	2	-
CO5:	3	1	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realiase the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, realize the operation of relay under pre-determined conditions.

I. Recommended Study Material:

Text Books:	
1.	Mehrdad Ehsani, Yimin Gao and Ali Emadi, "Modern Electric, Hybrid Electricand Fuel cell vehicles: Fundamentals, Theory and Design", CRC Press, 2004.
2.	Plug in Electric Vehicles in Smart Grids: Charging Strategy by Sumedha Rajakaruna, Farhad Shahnia, Arindam Ghosh, Springer.
3.	Iqbal Hussein, Electric and Hybrid Vehicles: Design Fundamentals, CRC Press, 2003.
Reference Books:	
1.	Deshang Sha, Guo Xu, HFrequency Isolated Bidirectional Dual Active Bridge DC–DC Converters with Wide Voltage Gain, Springer Nature Singapore Pte Ltd. 2019.
2.	Battery management systems, Volume I & II: Battery modeling. Artech

	House, 2015
3.	John Lowry and James Larminie , “Electric Vehicle Technology Explained”, Wiley
Web Material:	
1.	https://nptel.ac.in/courses/108/106/108106170/
2.	https://onlinecourses.nptel.ac.in/noc21_ee112/
3.	https://nptel.ac.in/courses/108/102/108102121/

EE491: ELECTRIC VEHICLES IN SMART GRID
7th Semester and 4th Year (Level IV)
B. Tech. (Electrical Engineering)
Minor Specialization Subject-4
Institute Elective

A. Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

B. Examination Scheme:

Theory Marks		Practical Marks		Total Marks
Internal	External	Internal	External	
30	70	25	25	150

C. Course Objectives:

As electrical power generation and economy is essential gradient for the industrial and all around development of any country, the objectives of the course are:

1. To understand the economics and effects on grid during charging.
2. To identify the impacts on system demand and on distribution system during different penetration level of EV charging.
3. To design control strategies for EVs to support frequency control of power system
4. To develop solutions based on ICT to support EV deployment.
5. To understand different protocol for grid interfacing and communication related to EVs.

D. Outline of the Course:

Sr. No.	Title of Units	Number of Hours
1	Introduction to EVs in Smart Grid	10
2	Influence Of EVs On Power System	15

3	Frequency Control Reserves & Voltage Support From EVs	17
4	ICT Solutions To Support EVs Deployment	12
5	Vehicular communication	06

Total hours (Theory) : 60 Hrs
Total hours (Lab) : 30 Hrs
Total hours : 90 Hrs

E. Detailed Syllabus:

1 Introduction to EVs in Smart Grid — 10 Hours 16.67%

Introduction, Impact of charging strategies, EV charging options and infrastructure, energy, economic and environmental considerations, V2G ,G2V and V2V Technology ,PEVs Charging Infrastructures ,Impact of EV charging on power grid, effect of EV charging on generation and load profile, Smart charging technologies, Impact on investment

2 INFLUENCE OF EVS ON POWER SYSTEM — 15 Hours 25%

Introduction, identification of EV demand, EV penetration level for different scenarios, classification based on penetration level, EV impacts on system demand: dumb charging, multiple tariff charging, smart charging, case studies, Effect of large scale EV charging on Distribution Systems.

3 FREQUENCY CONTROL RESERVES & VOLTAGE SUPPORT FROM EVS — 17 Hours 28.33%

Introduction, power system ancillary services, electric vehicles to support wind power integration, electric vehicle as frequency control reserves and tertiary reserves, voltage support and electric vehicle integration, properties of frequency regulation reserves, control strategies for EVs to support frequency regulation.

4 ICT SOLUTIONS TO SUPPORT EV DEPLOYMENT — 12 Hours 20%

Introduction, Architecture and model for smart grid & EV, ICT players in smart grid, smart metering, information & communication models, functional and logical models, technology and solution for smart grid: interoperability, communication technologies.

5 Vehicular communication 06 Hours 10%

Communication within vehicle, with grid, digital communication systems, vehicle network theory, embedded control, actuators, data analysis and importance of data in maintainability, concept of driverless car , NETWORKS AND PROTOCOLS : Overview of general-purpose networks and protocols -Ethernet, TCP, UDP, IP,ARP,RARP - LIN standard overview –workflow concept-applications –LIN protocol specification –signals - Frame transfer –Frame types –Schedule tables –Task behaviour model –Network management –status management - overview of CAN –fundamentals –Message transfer – frame types-Error handling –fault confinement-Bit time requirements.

F. Revised Bloom's Taxonomy

The below specification table shall be treated as a general guideline for students and teachers. The actual distribution of marks in the question paper may vary from the below table.

Level					
Remembrance	Understanding	Application	Analyze	Evaluate	Create
20	25	20	15	15	5

G. Course Outcomes (COs):

Upon successful completion of this course, a student would be able to

CO1:	Understand the economics and effects on grid during charging.
CO2:	Identify the impacts on system demand and on distribution system during different penetration level of EV charging.
CO3:	Design control strategies for EVs to support frequency control of power system
CO4:	Develop solutions based on ICT to support EV deployment.
CO5:	Understand different protocol for grid interfacing and communication related to EVs.

Mapping of course outcomes with program outcomes

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1:	3	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2:	3	2	2	1	1	-	-	-	-	-	-	-	-	-
CO3:	3	2	3	1	2	-	2	-	1	-	1	2	2	-
CO4:	3	3	3	2	2	-	2	-	1	-	1	2	2	-
CO5:	3	1	-	-	-	1	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation "-"

H. Instructional Methods and Pedagogy

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 10% component of the overall evaluation.
- Minimum two internal exams will be conducted and average of two will be considered as a part of 15% overall evaluation.
- Assignments/Surprise tests/Quizzes/Seminar/Tutorials based on course content will be given to the students for each unit/topic and will be evaluated at regular interval. It carries a weightage of 5% in the overall evaluation.
- The visit of CHARUSAT MEDICAL Campus HT control room and nearby substation will clear the concept of relay and relay setting and student will realize the actual application of relay.
- In lab session, the student will calculate the relay setting for particular application, will set the calculated value in relay and by performing the practical, and realize the operation of relay under pre-determined conditions.

I. Recommended Study Material:

Text Books:	
1.	SumedhaRajakaruna, FarhadShahnia and Arindam Ghosh, "Plug In Electric Vehicles inSmart Grids-Integration Techniques", Springer Science + Business Media Singapore PtLtd., 2015.
2.	Canbing Li, Yijia Cao, YonghongKuang and Bin Zhou, "Influences of Electric Vehicles on Power System and Key Technologies of Vehicle-to-Grid", Springer-Verlag Berlin Heidelberg, 2016.
3.	James Larminie and John Loury, "Electric Vehicle Technology – Explained", John Wiley & Sons Ltd, 2003
Reference Books:	
1.	Qiuwei Wu, "GRID INTEGRATION OF ELECTRIC VEHICLES IN OPEN ELECTRICITY MARKETS", John Wiley & Sons, Ltd, 2013.